



# Lamps: Learning Anatomy from Multiple Perspectives via Self-supervision in Chest Radiographs

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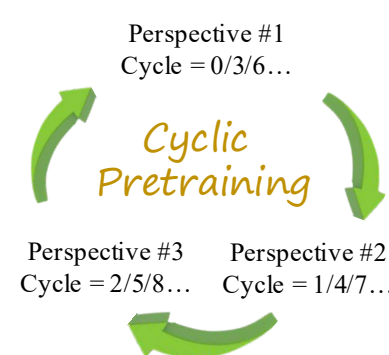
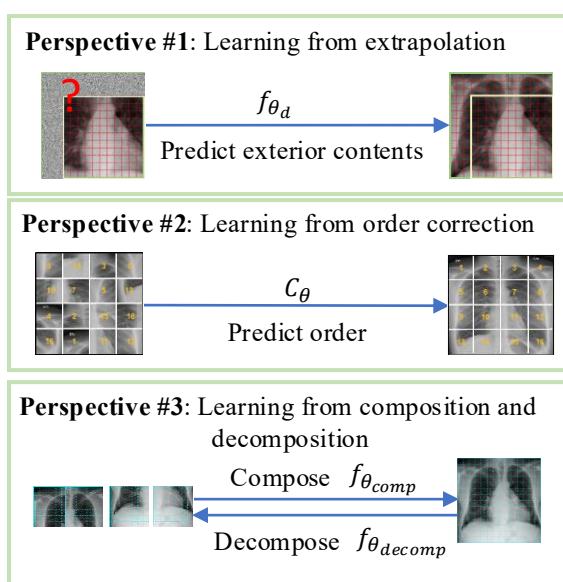
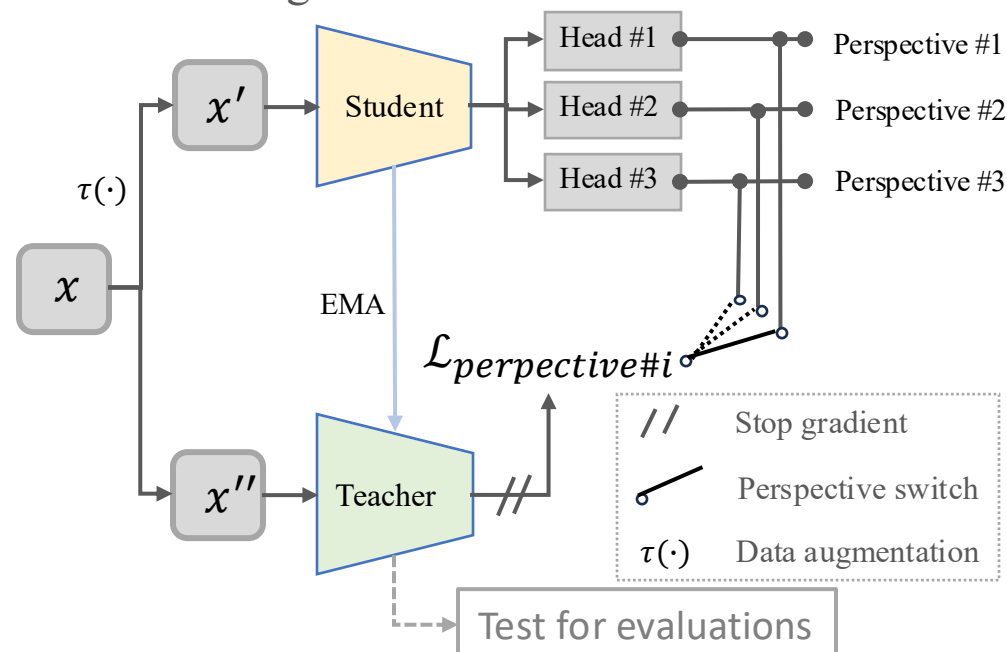


**Background:** Self-supervised learning (SSL) pretrains generic source models without using expert annotation, allowing the pretrained models to be uncovered of learned/emergent properties and quickly fine-tuned into high-performance application-specific target models and minimizing annotation cost.

**Motivation:** From an anatomical perspective, medical images acquired under the same imaging protocol exhibit **consistent spatial continuity**, **coherent structures**, and the **decomposability of organs and tissues into constituent parts**, owing to their stable relative positions. Leveraging these properties, we introduce a novel self-supervised learning approach, Lamps (Learning Anatomy from Multiple Perspectives).

**Contributions:** We propose Lamps, a novel SSL approach that learns from **anatomical continuity, coherence, and compositionality**, focusing on chest X-rays (CXR). Using large-scale unlabeled CXRs, Lamps (1) significantly outperforms existing SSL methods on **transfer learning**; (2) uncovers **emergent properties** of pretrained models, including the DNA-test, cross-patient anatomical correspondence, and anatomical locality; and (3) enables effective **large-scale SSL pretraining** for anatomical representation learning.

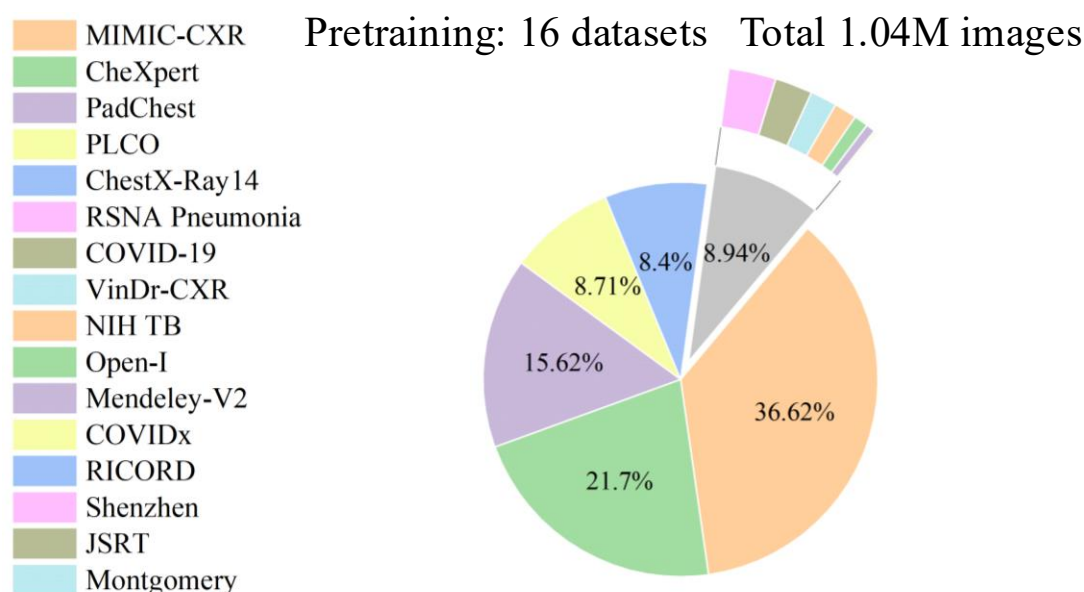
## A Pretraining



### Methodology Novelty

- Lamps collaboratively
- Learns **anatomy continuity** from extrapolation,
- Learns **anatomy layout** from order correction,
- Learns **hierarchies** from composition/decomposition by accruing knowledge from multiple perspectives via **cyclic pretraining**.

## B Chest X-ray Datasets

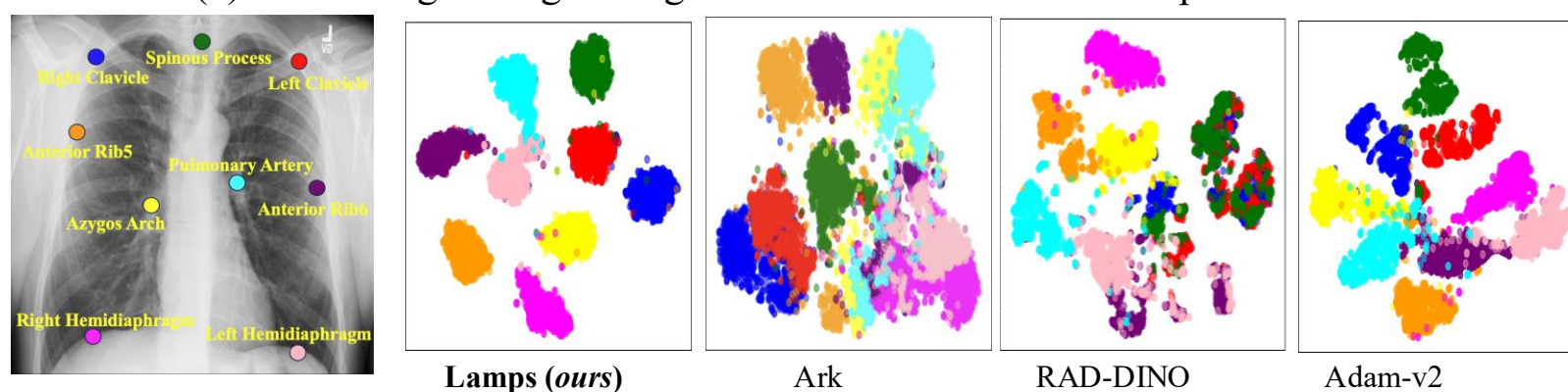


Evaluation: 10 datasets Total 388K images

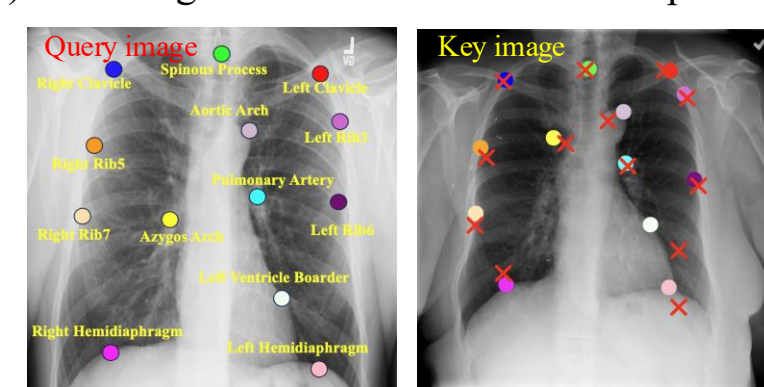
| Classification                                             | Segmentation                                            | Landmark correspondence                            | Report generation                                      |
|------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------|
| <b>ChestX-Ray14</b><br>112k images<br>14 labels            | <b>RSNA</b><br>26.7k images<br>3 labels                 | <b>CheXpert</b><br>223K images<br>14 labels        | <b>Shenzhen</b><br>662 images<br>2 labels              |
| <b>SIIM-ACR</b><br>12k images<br>Pneumothorax segmentation | <b>COVID-QU-EX</b><br>4.4k images<br>Covid segmentation | <b>JSRT</b><br>247 images<br>5 organs segmentation | <b>VinDr-Rib</b><br>245 images<br>20 ribs segmentation |
|                                                            |                                                         |                                                    | <b>Chest-Landmark</b><br>1k images<br>54 landmarks     |
|                                                            |                                                         |                                                    | <b>IU-XRay</b><br>7470 images<br>3955 reports          |

## C Evaluation tasks & Novel emergent and learned properties

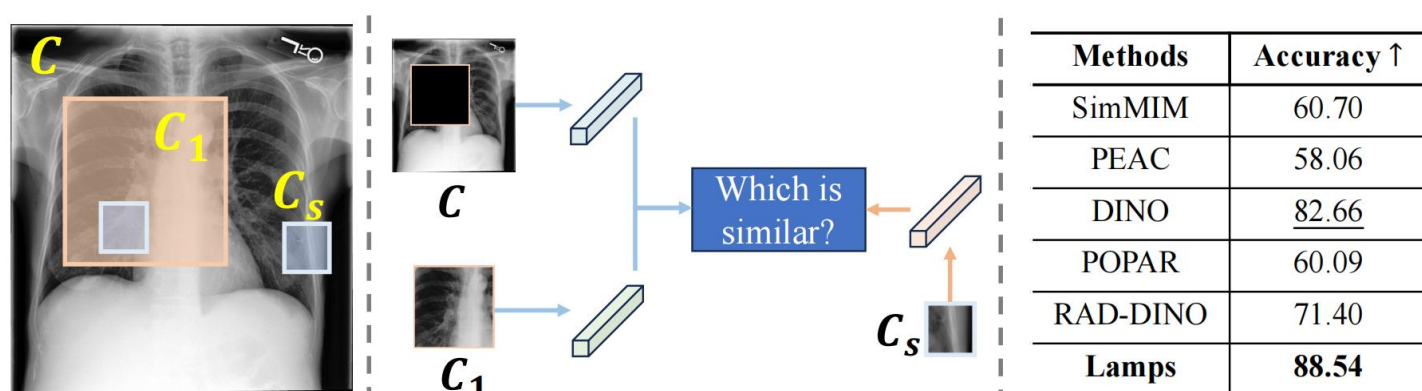
(a) Generating distinguishing anatomical structures across patients



(b) Matching identical anatomies across patients



(c) DNA-test: whether a part structure belongs to its corresponding whole



(d) Finetuning: achieves superior generalization

